Tagging: part of speech (POS), named entity recognition (NER)

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Applications of Natural Language Processing

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With slides by Brendan O'Connor

From bag of words to ordered structure

Sentence structure:

- word types such as noun, verb, …
- structure connecting words in the sentence: "the dog bit the mailman" is not the same as "the mailman bit the dog"

Why is this important?

My take on this: Preprocessing!

- Identifying events in news text: typically verbs
- Finding out "who does what to whom"
- In legal opinions, how much were particular defendants characterized as agents? Requires us to identify and trace mentions of those defendants in the text, and see what events they were involved in

Part of speech tags

- Syntax = how words compose to form larger meaning-bearing units
- parts of speech (POS) = syntactic categories for words
  - You can substitute words within a class and have a syntactically valid sentence.
  - Parts of speech determine how words can combine.

  - I saw the <u>dog</u>
  - I saw the <u>cat</u>
  - I saw the {<u>table</u>, <u>sky</u>, <u>dream</u>, <u>school</u>, <u>anger</u>, ...}

Demo

jupyter notebook: posdemo
available on Canvas under Lectures/8-tagging

Part of speech tagging

- I saw the fire today

- Fire!

Open and closed class words

- **Open class (lexical) words:**
  - Nouns:
    - Proper nouns: Alice, Italy
    - Common nouns: cat/cats, milk, …
  - (Main) verbs: see, prevaricated,…
  - Adjectives: loud, louder, loudest, yellow ,…
  - Adverbs: slowly, …
- **Closed class (functional) words:**
  - Determiners: the, a, …
  - Conjunctions: and, or, …
  - Prepositions: of, in, …
  - Pronouns: he, she, they, them, …
  - …

Do we want POS?

- Useful for many syntactic and other NLP tasks.
  - Phrase identification ("chunking")
  - Named entity recognition (proper nouns are often names)
  - Syntactic/semantic dependency parsing
  - Sentiment
- Either as features or heuristic filtering
- Esp. useful when not much training data
- Limitations
  - Coarse approximation of grammatical features
  - Sometimes cases are hard and ambiguous

POS patterns: simple noun phrases

POS patterns: simple noun phrases

- Quick and dirty noun phrase identification (Justeson and Katz 1995, Handler et al. 2016)
  - BaseNP = (Adj | Noun)* Noun
  - PP = Prep Det* BaseNP
  - NP = BaseNP PP*

More briefly, writing A for adjective, N for noun, P for preposition, a way to identify noun phrases
using **regular expressions over part of speech tags**:
((A | N)+ | ((A | N)* (N P)?) (A | N)*) N

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Examples from Jurafsky and Martin:

A N - linear function

N N - regression coefficients

A A N - Gaussian random variable

A N N - cumulative distribution function

N A N - mean square error

N N N - class probability function

N P N - degrees of freedom

POS patterns: sentiment

Turney (2002): identify bigram phrases, from unlabeled corpus, useful for sentiment analysis.

first word	Second word	Third word (not extracted)
JJ	NN or NNS	anything
RB, RBR, RBS	JJ	not NN nor NNS
JJ	JJ	not NN nor NNS
RB, RBR, RBS	VB, VBD, VBN, VGG	anything

phrases include:
online experience, low fees, local branch, small part, inconveniently located, other bank…

POS Taggers

- How do you predict POS tags?
- Off-the-shelf models widely available, at least for mainstream varieties of major world languages
  - e.g. Spacy, Stanza, CoreNLP, NLTK, etc.
- Typically use logistic regression-like models
  - Each token instance is a classification problem (but: see next slide!)
  - Labeled datasets: e.g. https://universaldependencies.org/

POS Tagging: lexical ambiguity

Can we just use a tag dictionary
(one tag per word type)?

Types:		**WSJ**		**Brown**	
Unambiguous	(1 tag)	44,432	(**86%**)	45,799	(**85%**)
Ambiguous	(2+ tags)	7,025	(**14%**)	8,050	(**15%**)

Tokens:					
Unambiguous	(1 tag)	577,421	(**45%**)	384,349	(**33%**)
Ambiguous	(2+ tags)	711,780	(**55%**)	786,646	(**67%**)

Most words types are unambiguous ...

But not so for *tokens*!

- Ambiguous word types tend to be the common ones.
  - I know **that** he is honest = IN (relativizer)
  - Yes, **that** play was nice = DT (determiner)
  - You can't go **that** far = RB (adverb)
- Tagging: **Sequence tagger,** takes previous tokens into account

POS Tagging: baseline

- Baseline: each word type's most frequent tag. >90% accuracy!
  - Simple baselines are very important to run!

- Though...
  - I get 91.8% accuracy for token tagging
  - ...but, 18.6% whole-sentence accuracy (!)

- Many other NLP tasks can be cast as tagging
  - Example: Named entity recognition

Named entity recognition

SOCCER - [PER BLINKER] BAN LIFTED . [LOC LONDON] 1996-12-06 [MISC Dutch] forward [PER Reggie Blinker] had his indefinite suspension lifted by [ORG FIFA] on Friday and was set to make his [ORG Sheffield Wednesday] comeback against [ORG Liverpool] on Saturday . [PER Blinker] missed his club's last two games after [ORG FIFA] slapped a worldwide ban on him for appearing to sign contracts for both [ORG Wednesday] and [ORG Udinese] while he was playing for [ORG Feyenoord].

Figure 1: Example illustrating challenges in NER.

- Goal: for a fixed entity type inventory (e.g. PERSON, LOCATION, ORGANIZATION), identify all *spans* from a document
  - Name structure typically defined as flat (is this good?)

[Ratinov and Roth 2009]

BIO tagging

[$_{PER}$ **Jane Villanueva**] of [$_{ORG}$ **United**] , a unit of [$_{ORG}$ **United Airlines Holding**] , said the fare applies to the [$_{LOC}$ **Chicago**] route.

- Can we map identify phrases (spans) identification to token-level tagging?

BIO tagging

[$_{PER}$ **Jane Villanueva**] of [$_{ORG}$ **United**] , a unit of [$_{ORG}$ **United Airlines Holding**] , said the fare applies to the [$_{LOC}$ **Chicago**] route.

Words	IO Label
Jane	I-PER
Villanueva	I-PER
of	O
United	I-ORG
Airlines	I-ORG
Holding	I-ORG
discussed	O
the	O
Chicago	I-LOC
route	O
.	O

Figure 17.7 NER as a sequence model,

"IO" tagging: issues?

BIO tagging

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the	O	O
Chicago	I-LOC	B-LOC
route	O	O
.	O	O

BIO is a lossless representation of flat spans
Easy to extract spans from tagger output